

MySQL Capstone Project

Hospital Appointment and Billing Analytics

Database Design:

- Developed a Hospital Appointment and Billing Analytics database using MySQL.
- Created 7 tables:
 1. Patients
 2. Departments
 3. Doctors
 4. Appointments
 5. Treatments
 6. Bills
 7. Payments
- Used primary keys to uniquely identify records in each table.
- Designed the database to support hospital operations and management reporting.

Table Relationships:

- **Departments → Doctors** : One department can have many doctors.
- **Patients → Appointments** : One patient can book multiple appointments.
- **Doctors → Appointments** : One doctor can attend multiple appointments.
- **Appointments → Treatments** : Each appointment may have a treatment record.
- **Appointments → Bills** : Each appointment generates a bill.
- **Bills → Payments** : Bills are linked to payment records for tracking collections.
- Relationships were implemented using foreign keys to maintain data consistency.

Key Insights:

- The database successfully integrates patient, doctor, appointment, treatment, billing, and payment information.
- Appointment reports provide a complete view of patient visits and treatment details.
- Billing and payment reports help monitor hospital revenue and outstanding balances.

- Department-wise and doctor-wise analysis helps evaluate hospital performance and workload distribution.
- Data quality checks help identify missing, invalid, or inconsistent records.
- The generated reports provide valuable insights for operational monitoring, financial tracking, and resource planning.